

Centennial Jing-Zhang AI Belt · Design Booklet (A3)

THE EMERGENT BELT · JINGZHANG HYPER LINE

Title: THE EMERGENT BELT — Jing-Zhang Hyper Line

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Method: CASA complexity urban science × Amap POI 25,476 pts + OSM fabric

Discipline: all numbers computed locally (EPSG:4548), single-source registry

Caveat: provisional boundary for generation/display/self-check only; regulatory indicators pending; conceptual recommendations

Fig.1 · Master Plan: one line, three folds, two wings, seven nodes (concept)



Fig.1 · Master plan (real fabric base + concept overlay)

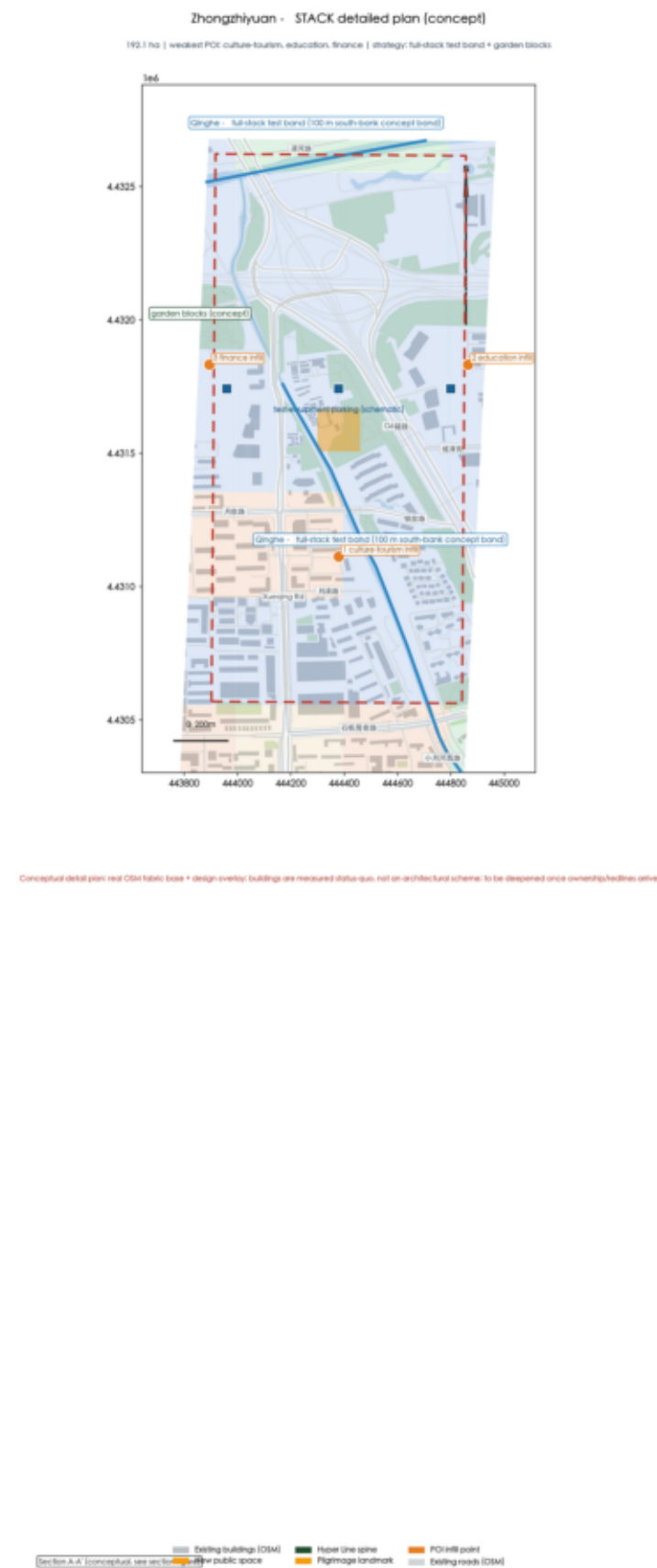


Fig.2 · Zhongzhiyuan · STACK detailed plan

Origin · ORIGIN detailed plan (concept)

104.3 ha | integration 93.5% (a century-old siting still the structural core) | medical density 2.88/km², the lowest → infill healthcare, talent apartments, community services



Fig.3 · Origin · ORIGIN detailed plan

Dazhongsi · FRONT detailed plan (concept)

72.0 ha | syntactic percentile 20.2% (southern periphery) | POI 511/km², the densest of the three · jobs-residence 0.344 | infill finance, culture-tourism, research

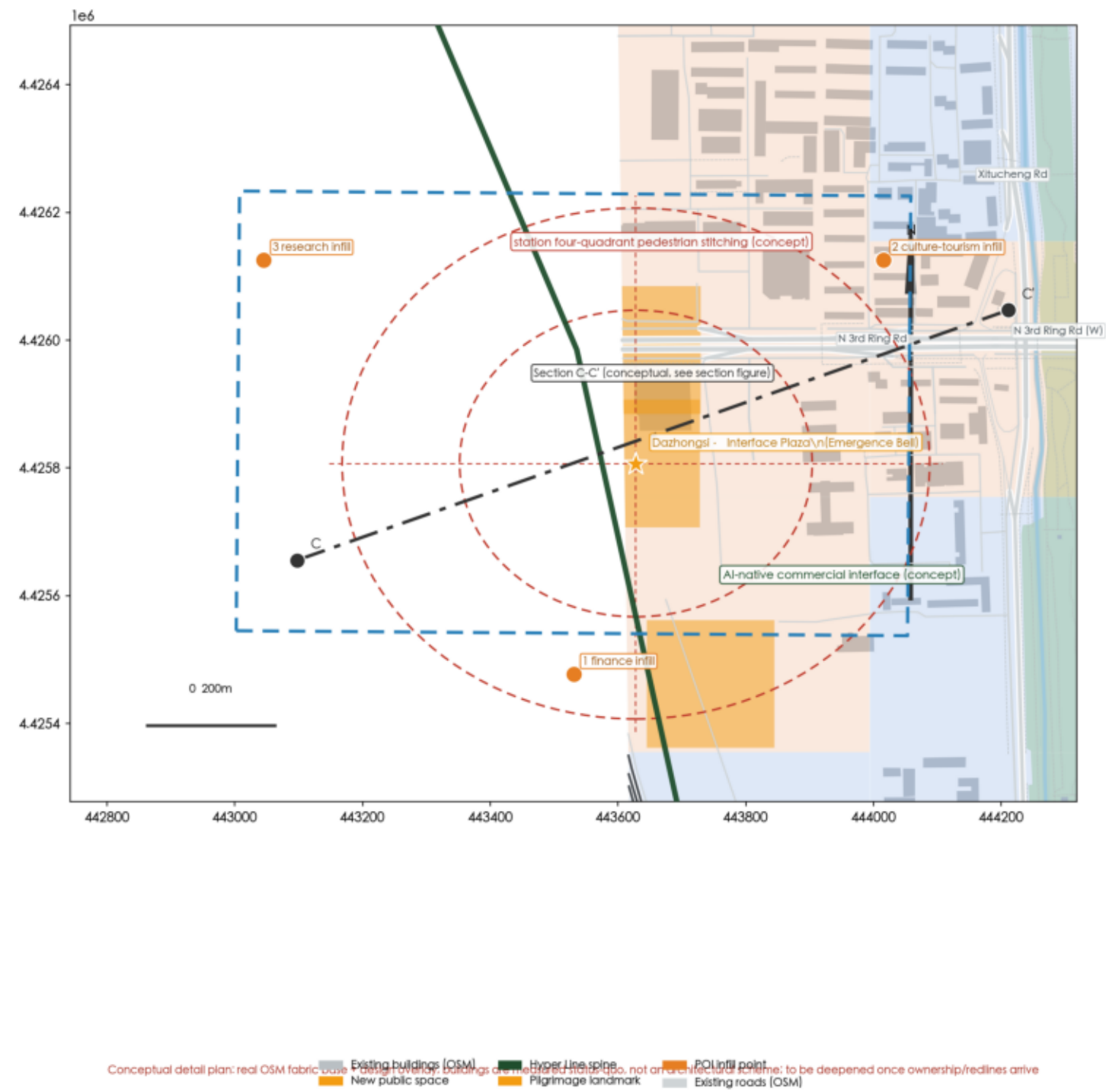


Fig.4 · Dazhongsi · FRONT detailed plan

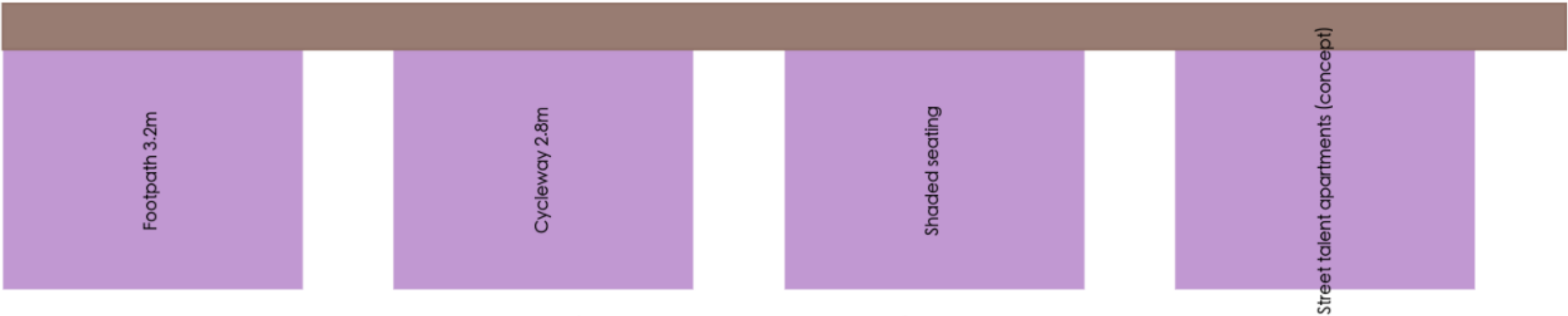
Zhongzhiyuan · HYPER STACK section: garden test band (concept) (approx. 21m, conceptual section)



Conceptual section, not an engineered one; dimensions are design intent (pending road redline and heritage review)

Fig.5 · Zhongzhiyuan section (concept)

Origin Community · HYPER ORIGIN section: near-campus street (concept) (approx. 21m, conceptual section)



Conceptual section, not an engineered one; dimensions are design intent (pending road redline and heritage review)

Fig.6 · Origin section (concept)

Dazhongsi · HYPER FRONT section: interface street (concept) (approx. 21m, conceptual section)



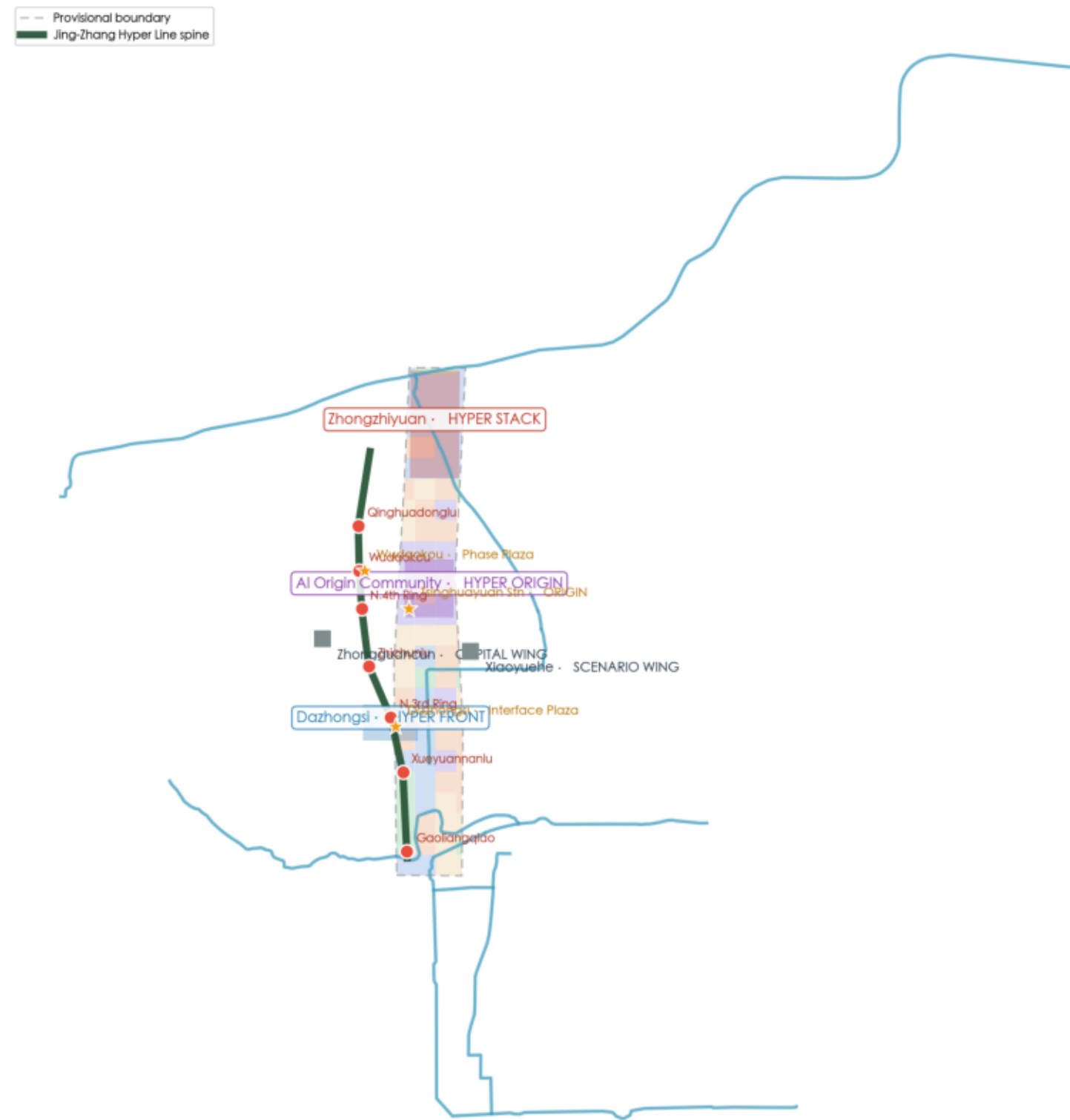
Conceptual section, not an engineered one; dimensions are design intent (pending road redline and heritage review)

Fig.7 · Dazhongsi section (concept)



Fig.8 · Vision aerial 2035 (AI concept render)

Fig.1 · Belt overview: one line, three folds, two wings, seven nodes



Sources: official announcement + OSM + Amap POI [2026-08-15]; provisional boundary dashed only, not an official redline; all conceptual

Fig.9 · Belt overview

Fig.2 · Three-level scope and land-use structure

Fig.2a · Land-use structure & POI-driven zoning (concept)

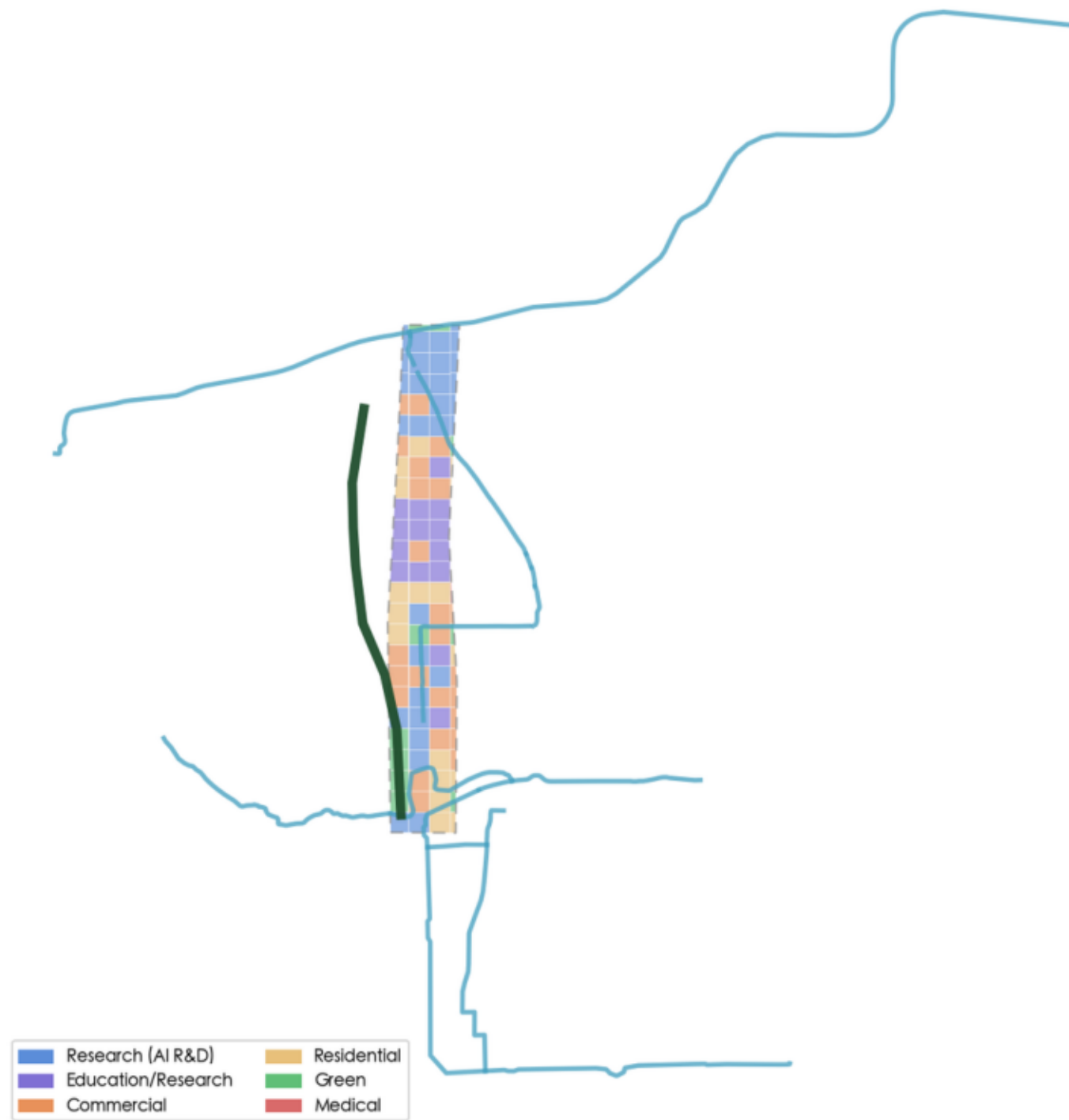


Fig.2b · Land-use mix (10k m², conceptual)

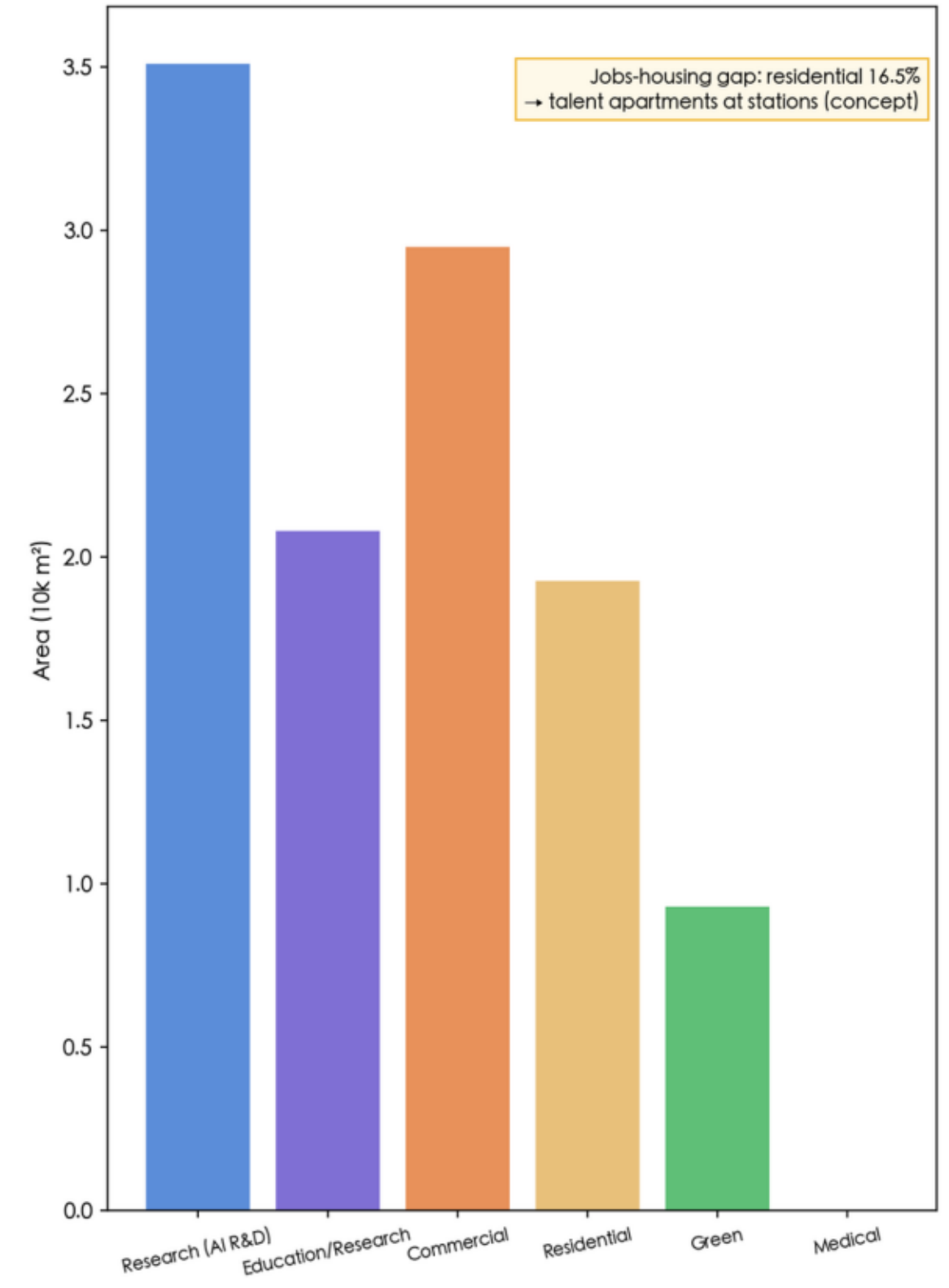
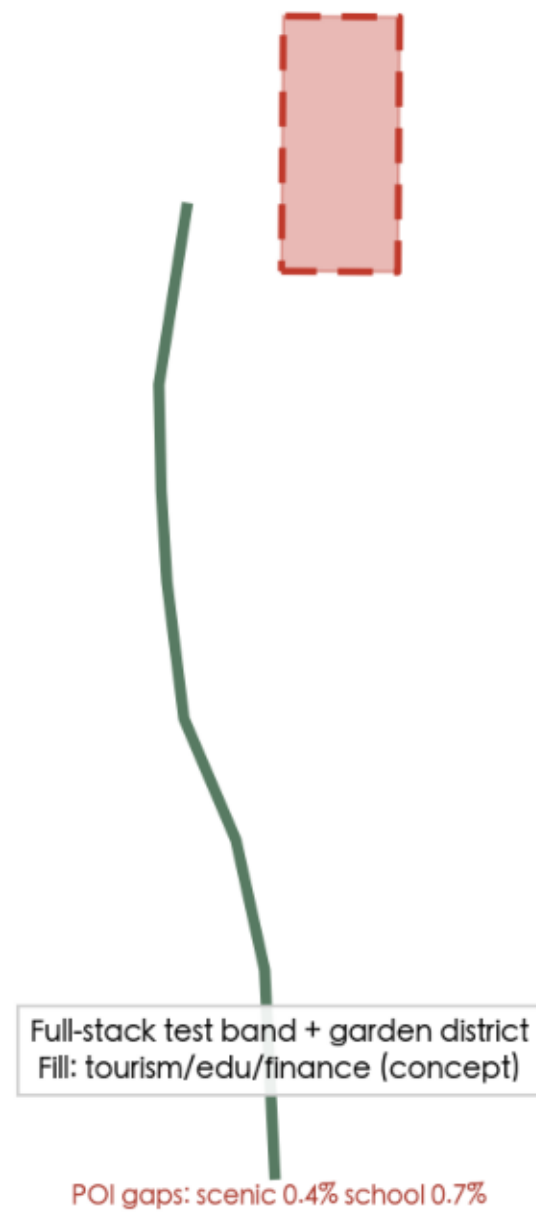


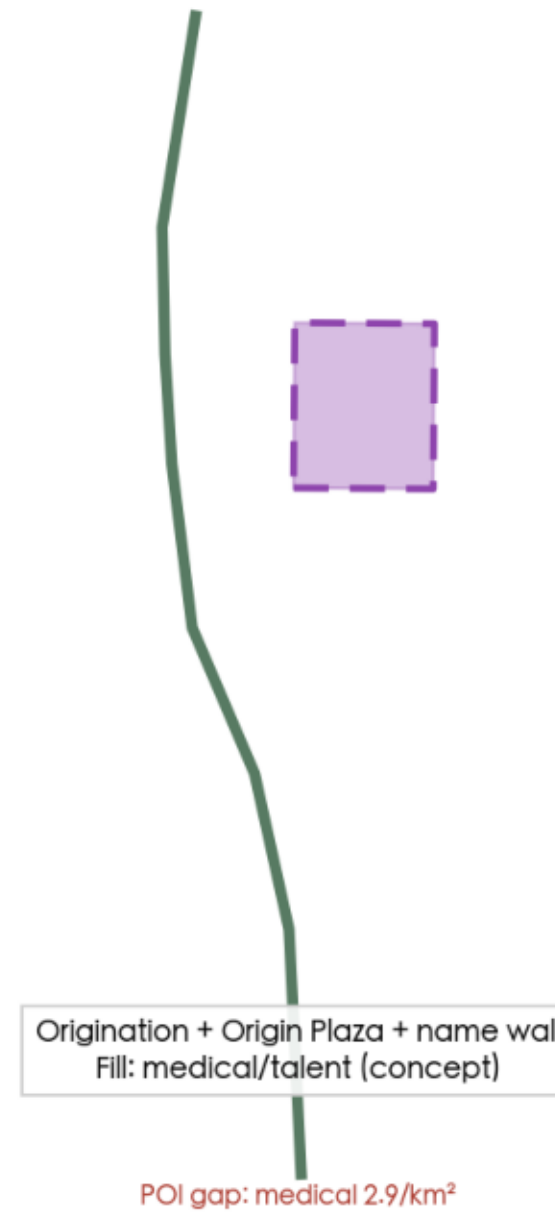
Fig.10 · Land use & POI zoning

Fig.3 · Key-area detailed design positions

Zhongzhiyuan · HYPER STACK



Origin Community · HYPER ORIGIN



Dazhongsi · HYPER FRONT

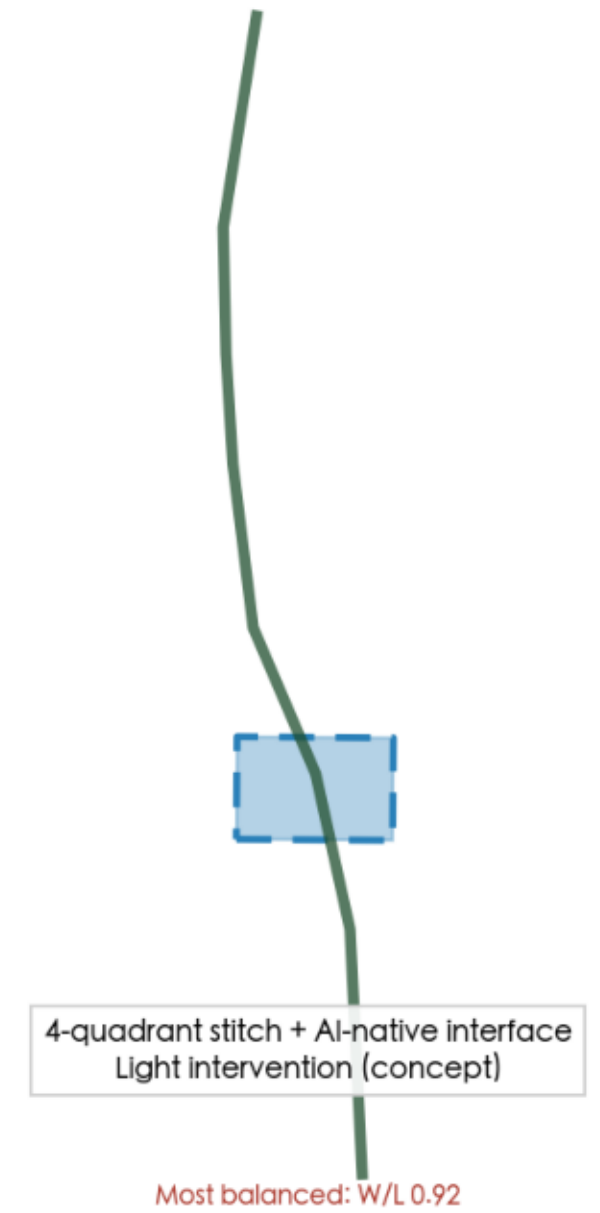


Fig.11 · Key-area positions (dashed = provisional)



Fig.12 · Seven-node stitching & blue-green

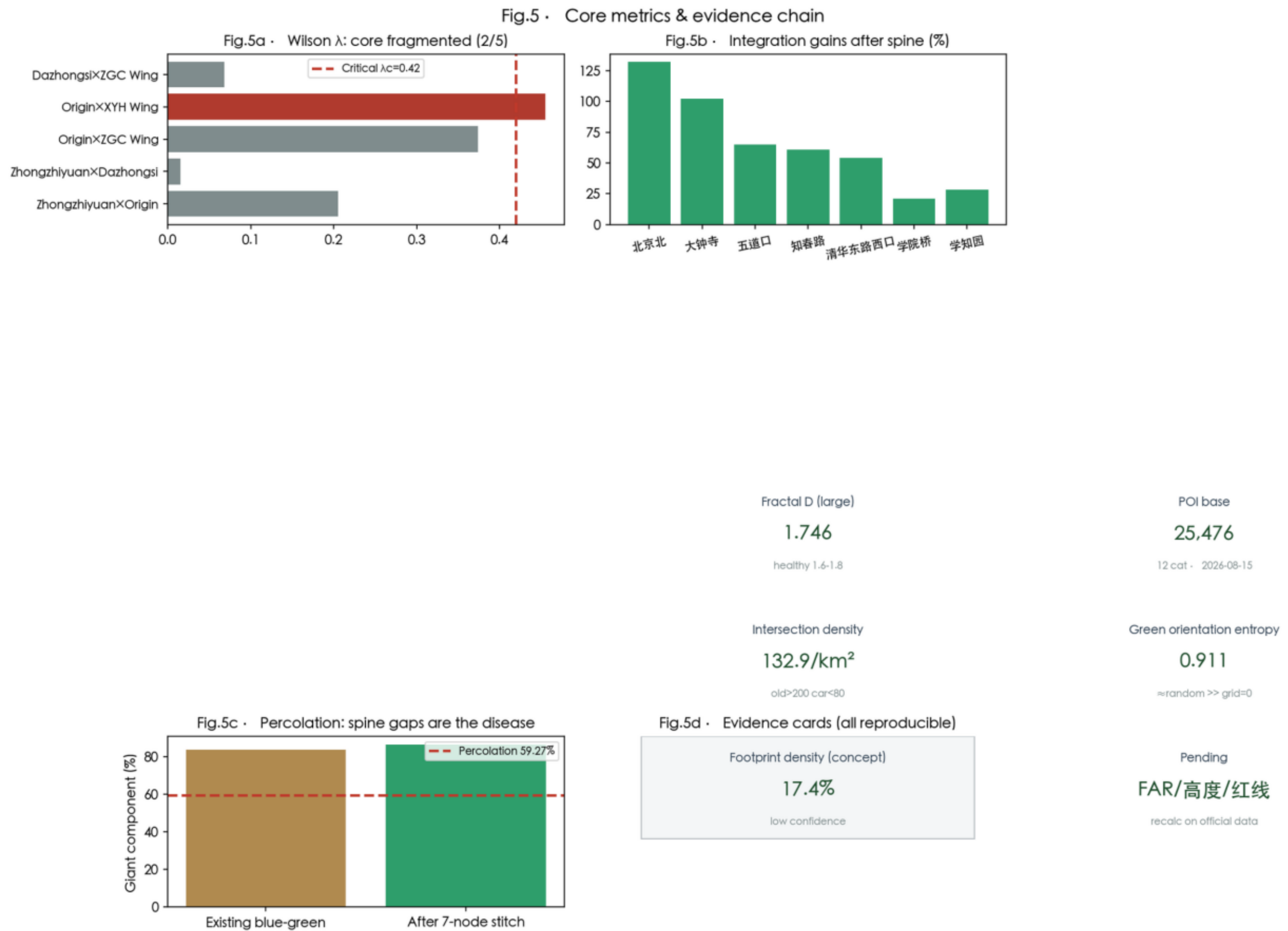
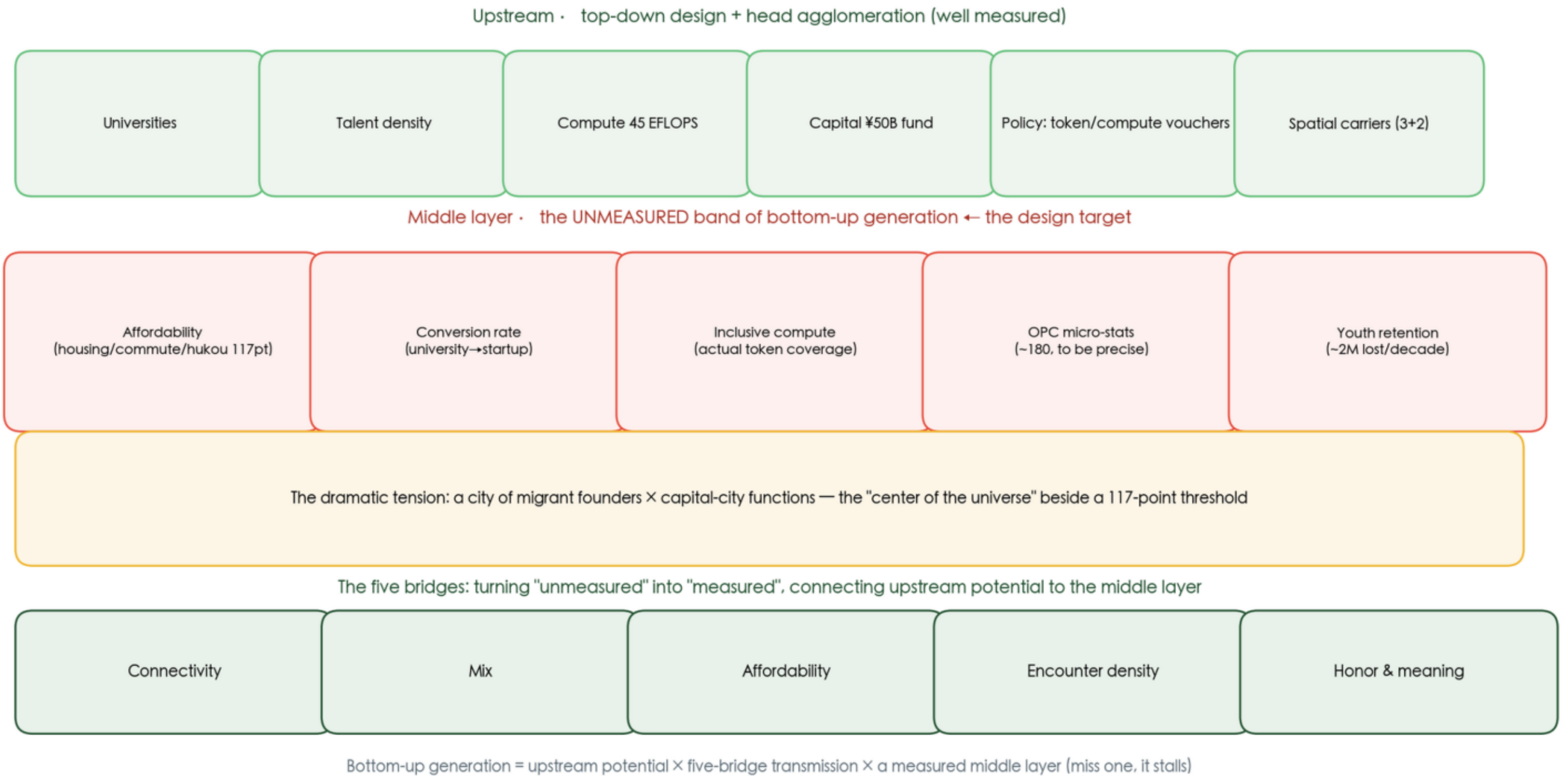


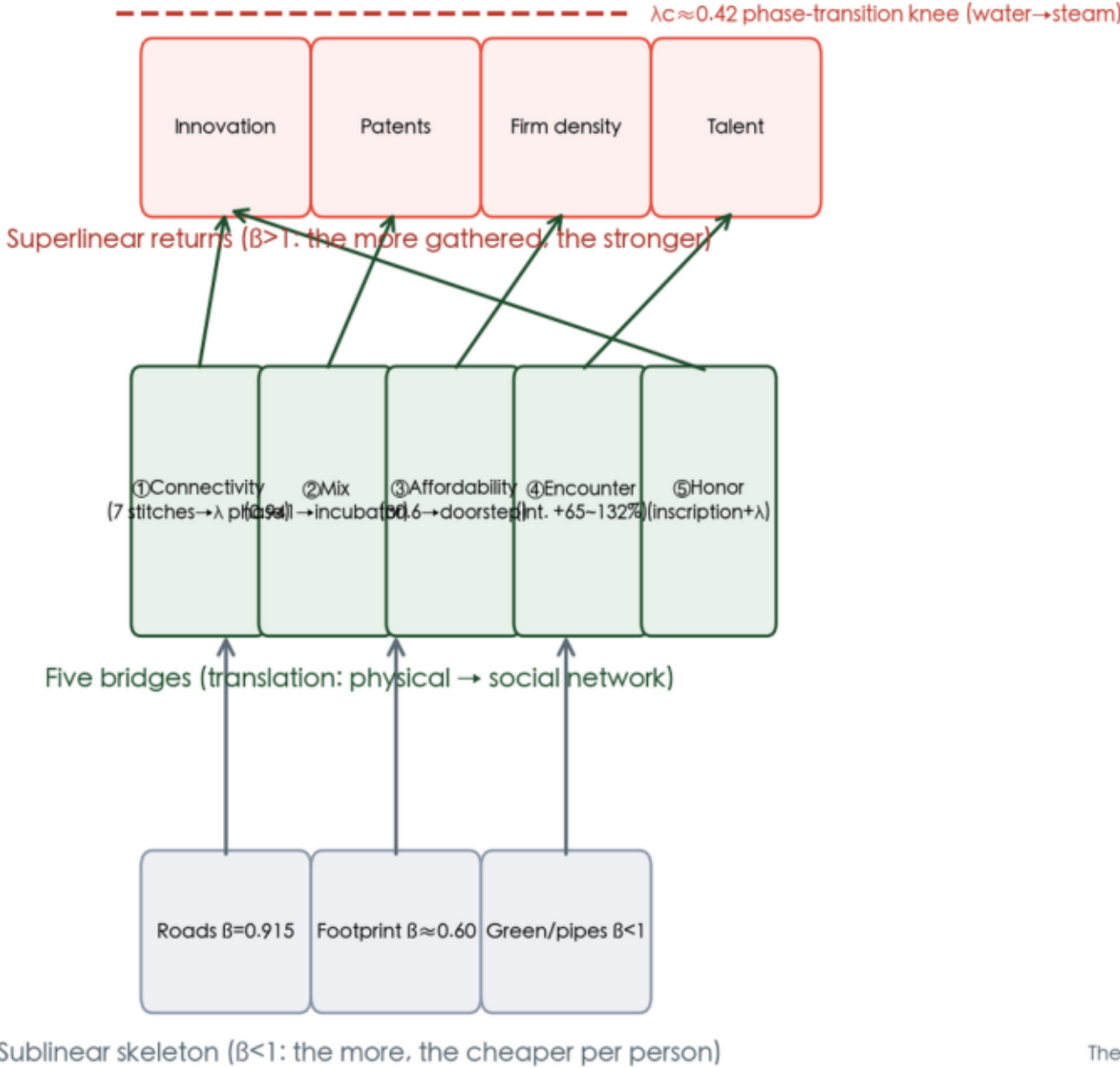
Fig.13 · Core metrics & evidence



Sources: ~60 public records (see sources.json ecosystem entry); red = variables lacking official measurement

Fig.14 · Ecosystem map

Fig. · The sublinear→superlinear transmission map (the scientific heart)



"Bend the curve toward every citizen"

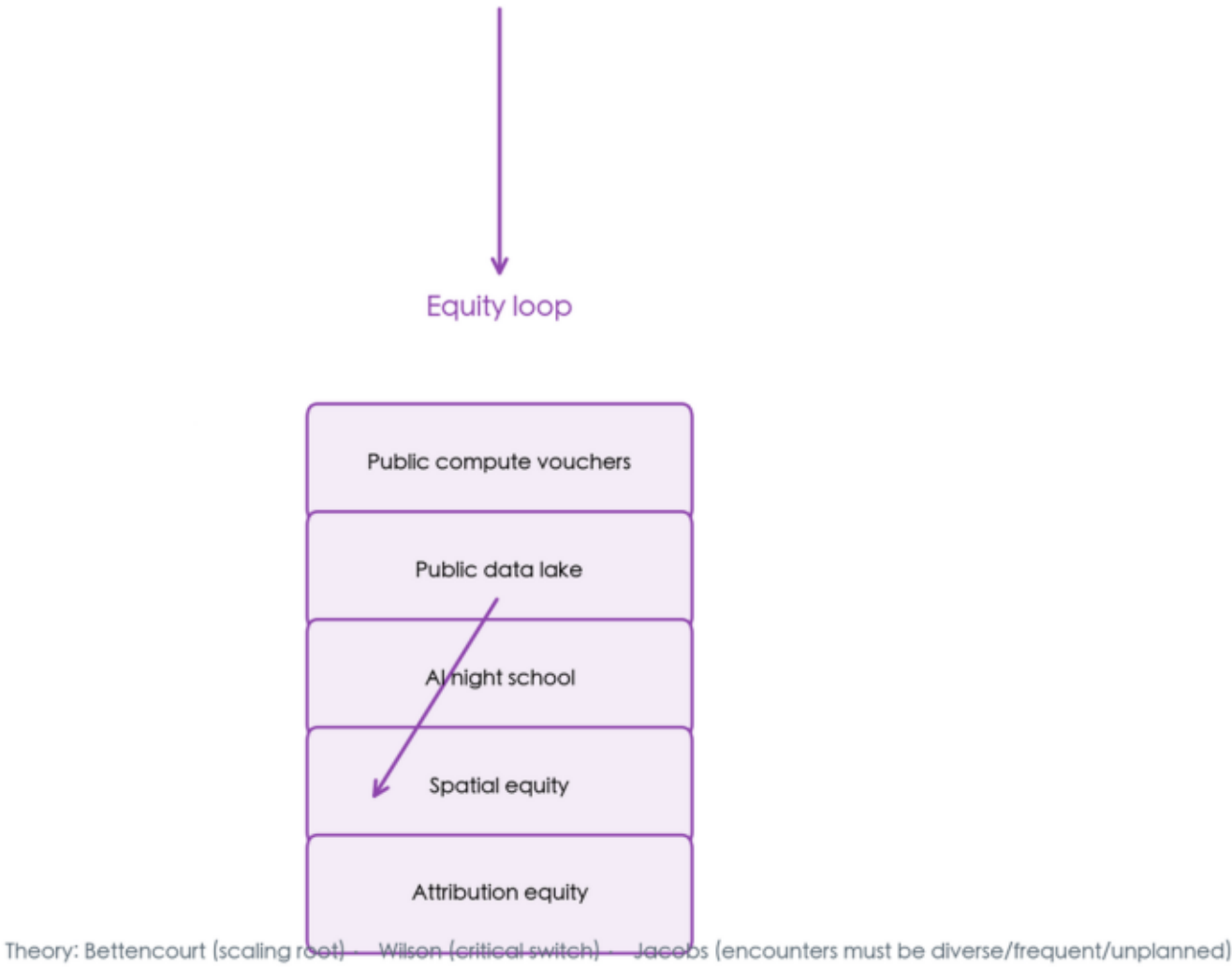


Fig.15 · Transmission map (five bridges + equity loop)

Fig. C8-C10 · Scaling deviation / employment balance / health factor (threshold system)

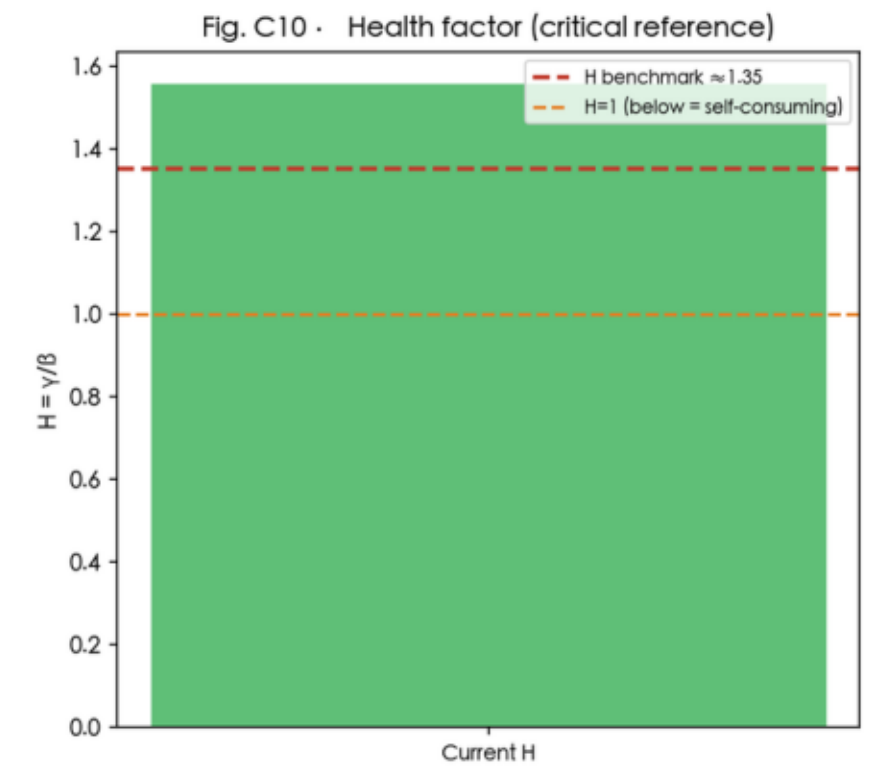
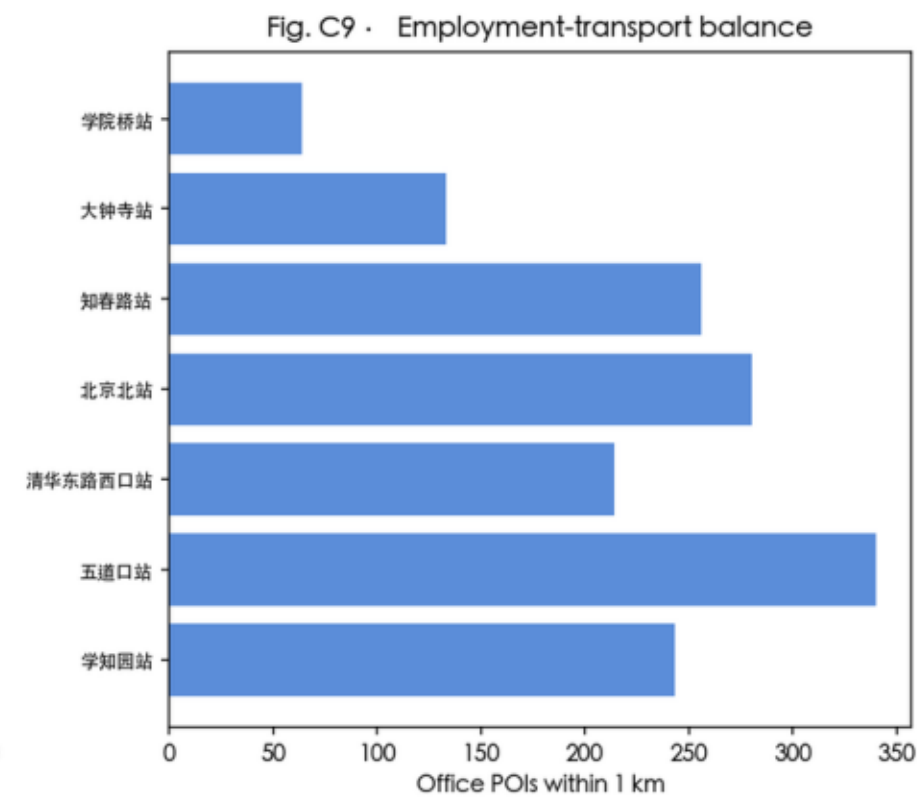
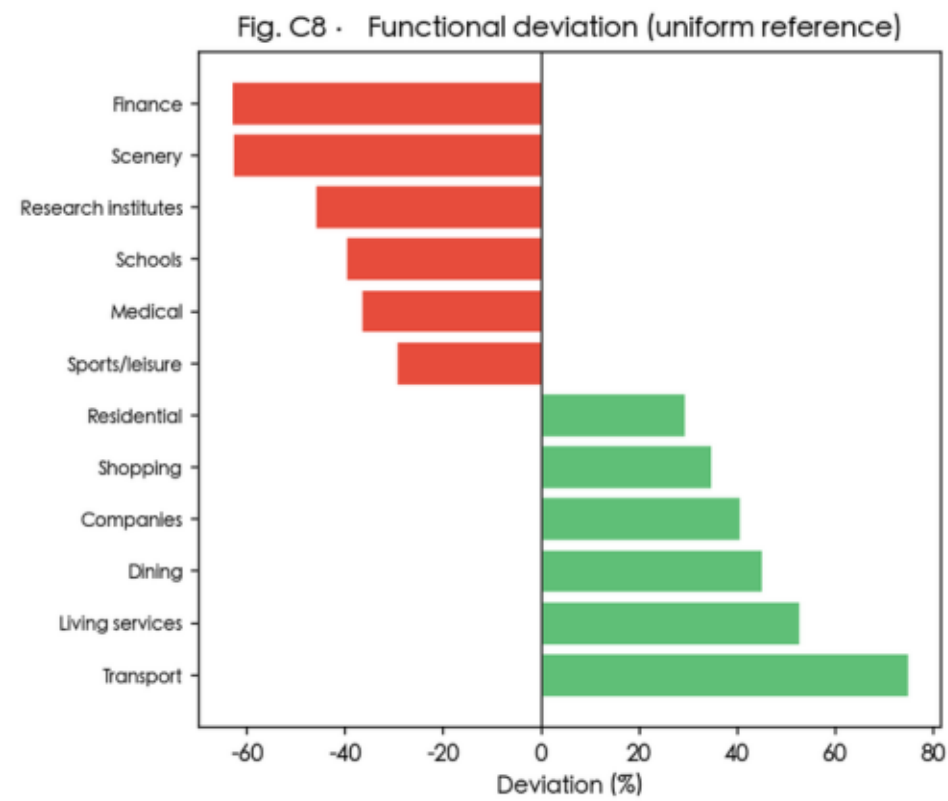


Fig.16 · Deviation / employment balance / health factor

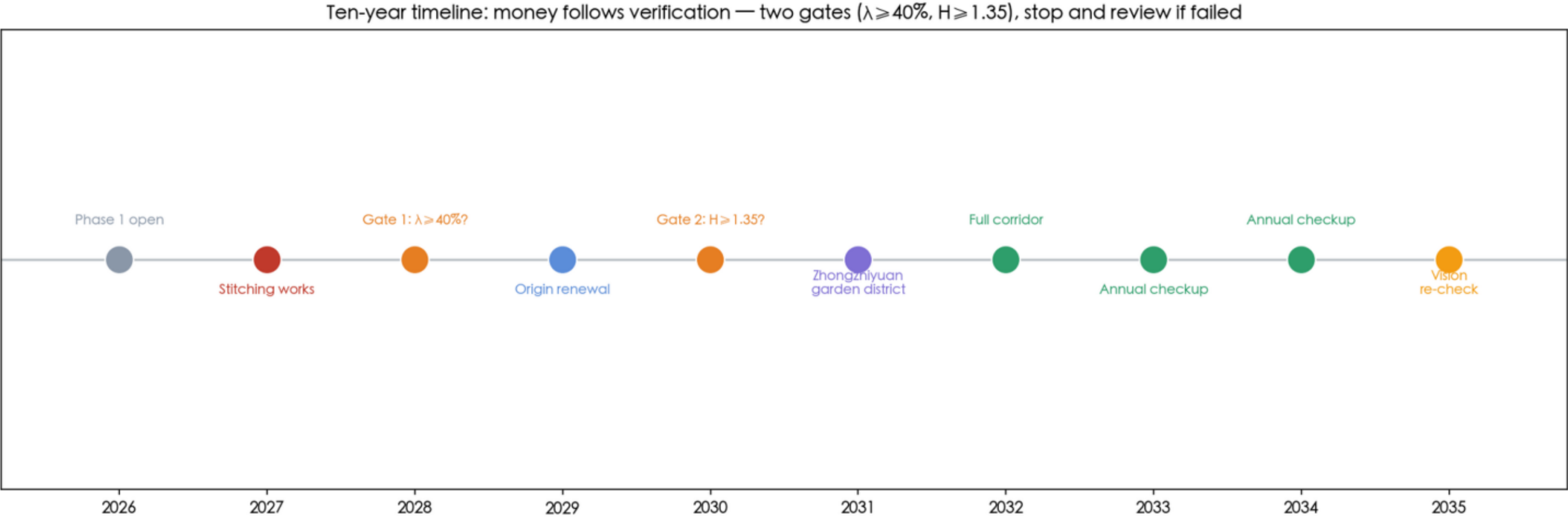


Fig.17 · Operations calendar & milestones (concept)

Conceptual height zoning (schematic): bands step down toward the spine, low front / high back

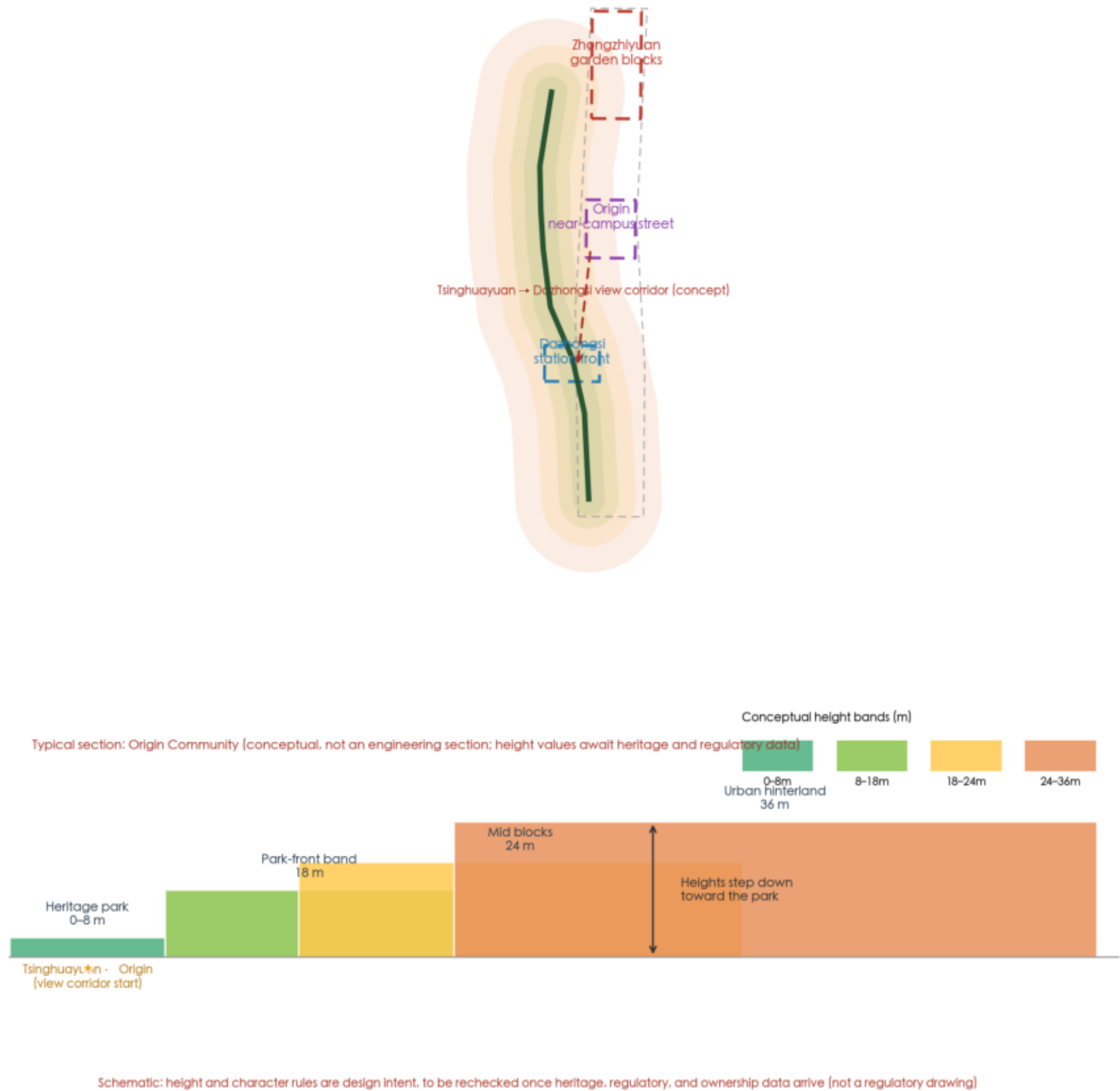


Fig.18 · Character & conceptual height zoning (schematic)

Fig C18 - Evidence for A-BOUNDARY-003: rough substitute boundary vs name semantics (working boundary still provisional)

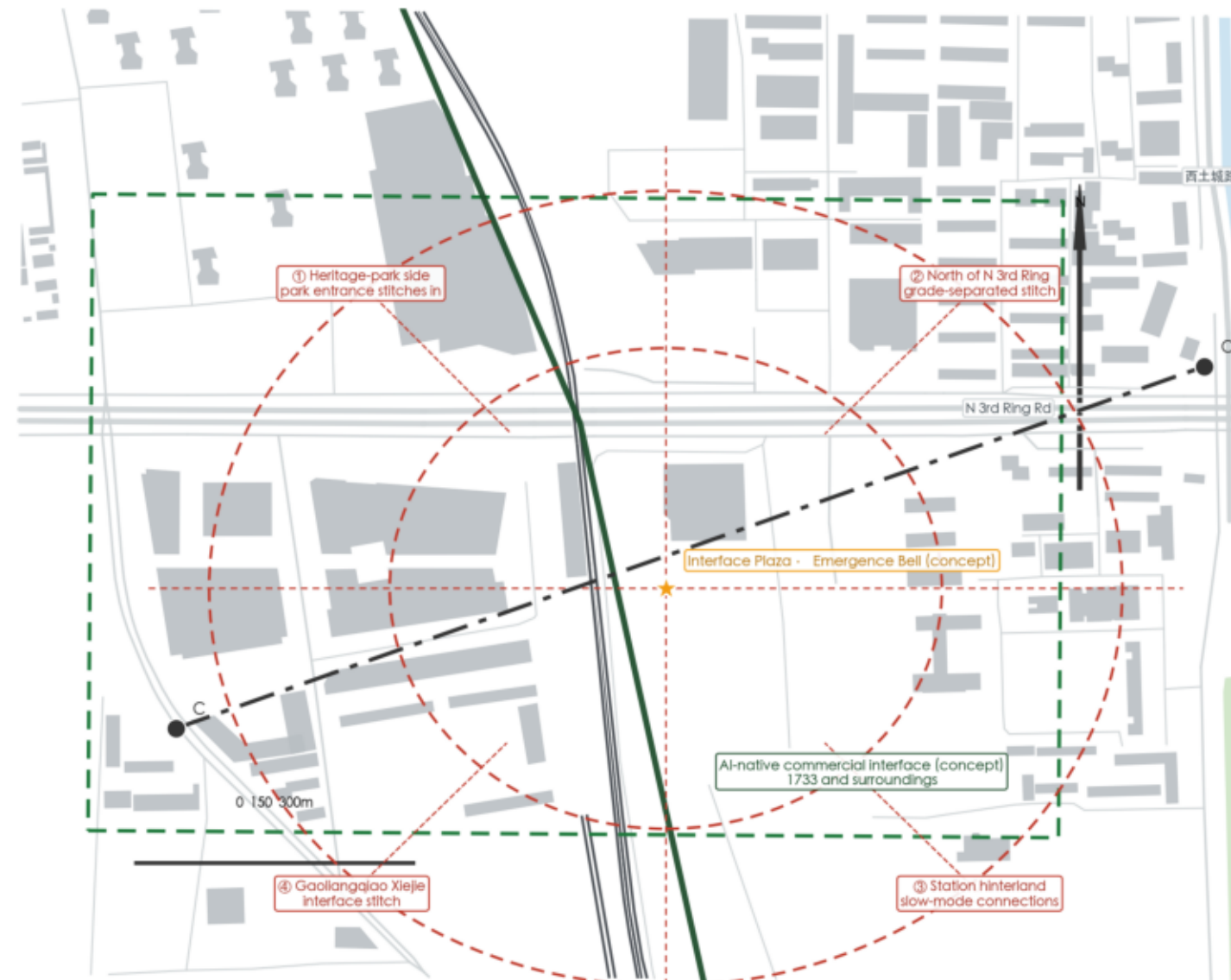


Brief evidence: ① provisional_boundaries_base.md 136 — "Dazhongzi AI Industry Cluster" | the brief rough boundary PROV-KEY-003 is a rough substitute boundary for the Dazhongzi Station | 72.3 ha — yet the supplied PROV-KEY-003 polygon actually sits near Beijing North Station [39 944-39 950], ~2 km from the named Dazhongzi Station [39 942-39 948]. ② missing-data.md 18 — the precise polygons of the three key areas are missing. This proposal corrects the working boundary by name semantics (assumption A-BOUNDARY-003); a full-package recalculation follows once the official polygon arrives.

Base: OSM ID OpenStreetMap contributors, CC-BY; all boundaries provisional, not official redlines

Fig.19 · Evidence page for A-BOUNDARY-003

Fig C19 · Dazhongsi station-district four-quadrant stitching, enlarged (conceptual design)



Conceptual schematic: the stitch modes (grade-separation / slow-mode links) are conceptual suggestions, to be deepened once transport, municipal, and ownership data arrive; section C-C' shown in the detailed section figure.

Base: OSM [© OpenStreetMap contributors, ODbL]

Fig.20 · Dazhongsi four-quadrant stitching, enlarged (concept)

Design task responses

THE EMERGENT BELT · JINGZHANG HYPER LINE

- agent.1 Naming/logo: THE EMERGENT BELT; HYPER STACK/ORIGIN/FRONT; logo = the superlinear line
- agent.2 Ecosystem cases (7): King's Cross / Kendall Sq / Stanford / Yuehai / Shibuya / Cambridge / Rothschild
- agent.3 10 scenario cards (4 industrial tests) + 5 personas
- agent.4 Three pilgrimage landmarks: Origin / Phase Plaza / Interface Plaza
- agent.5 Narrative: 1909 Y-switch self-built railway → 2026 superlinear AI agglomeration
- agent.6 Operations: dev festival / phase forum / open test seasons / honor wall (conceptual)

Quantified diagnosis & design targets

THE EMERGENT BELT · JINGZHANG HYPER LINE

- Wilson λ : core 2/5 connected; Zhongzhiyuan×Dazhongsi $\lambda=0.033$ deeply subcritical
- 7 spine gaps → stitching = phase transition
- Space syntax: +70% Dazhongsi, +49% Beijing North, +43% Wudaokou, +24% Zhichunlu
- Fractal D=1.746 (healthy 1.6–1.8); intersections 132.9/km²; POI 25,476
- Jobs-housing gap: residential 16.5% → talent apartments (concept)
- POI gaps: Zhongzhiyuan tourism/edu/finance; Origin medical; Dazhongsi research/edu

Phasing & risk (conceptual)

THE EMERGENT BELT · JINGZHANG HYPER LINE

- Near (1–3y): spine stitching + Dazhongsi; mid (3–5y): Origin; long (5–10y): Zhongzhiyuan
- Pending: official redlines/regulatory/heritage/inventory/pipelines → full recalculation
- Rights: locally generated; naming needs trademark/font clearance; no PII or unauthorized assets
- Boundary: does not replace formal planning; no confirmed government arrangements